

B747 Ramp Radial Magnet Cosmetic Standard-A3

Rev. A3
Magnet Vendor: Magsound
2025/08/04

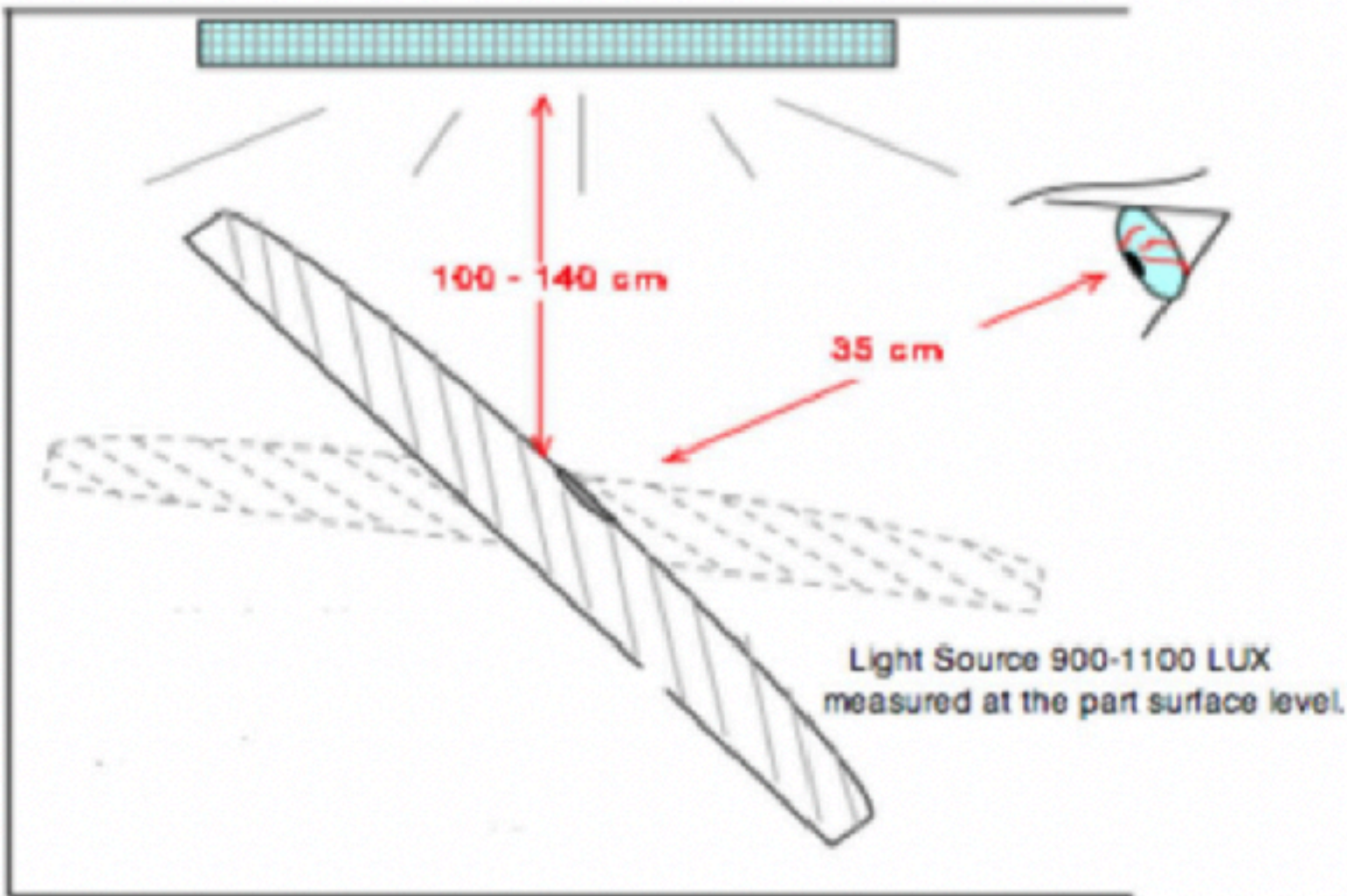
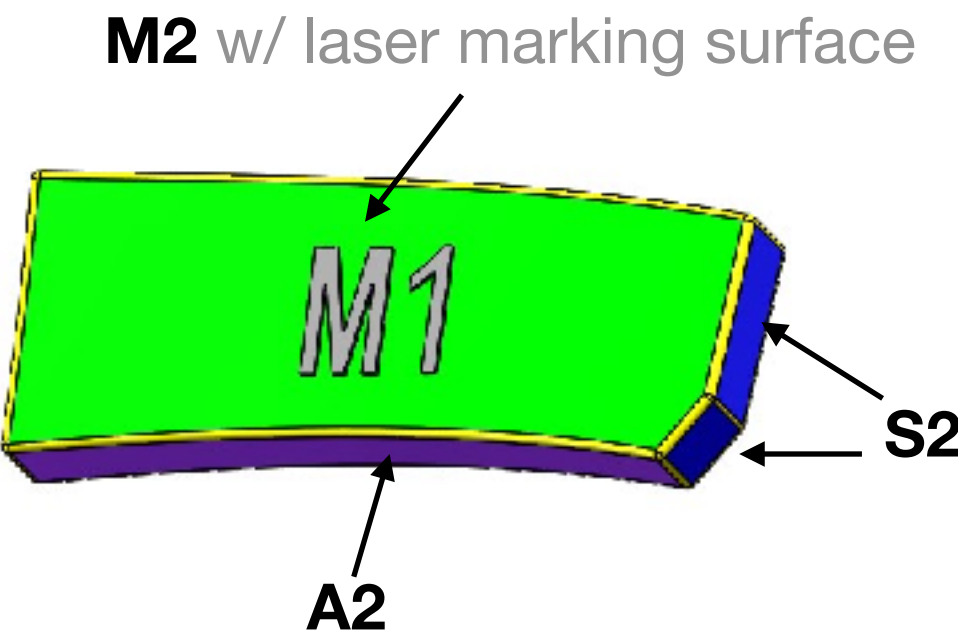
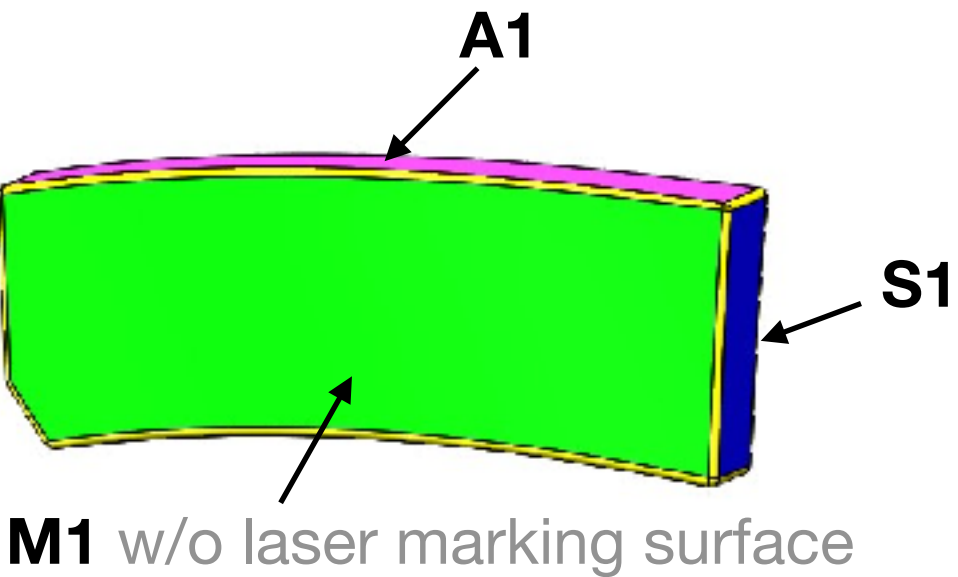


Fig. 1 Inspection criteria for opaque surfaces / all classes

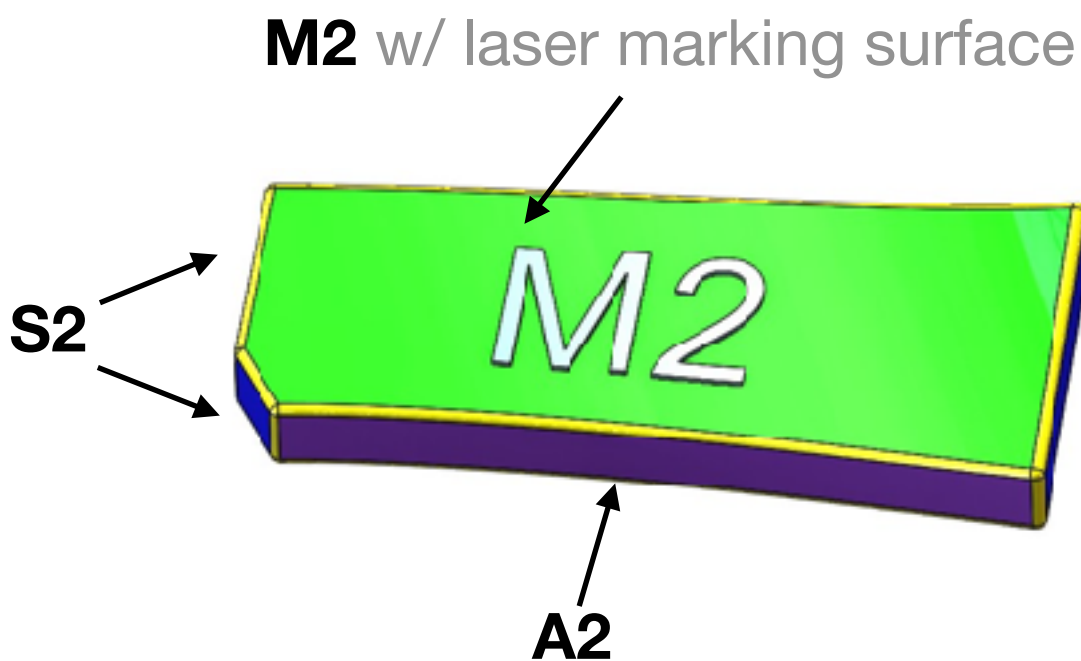
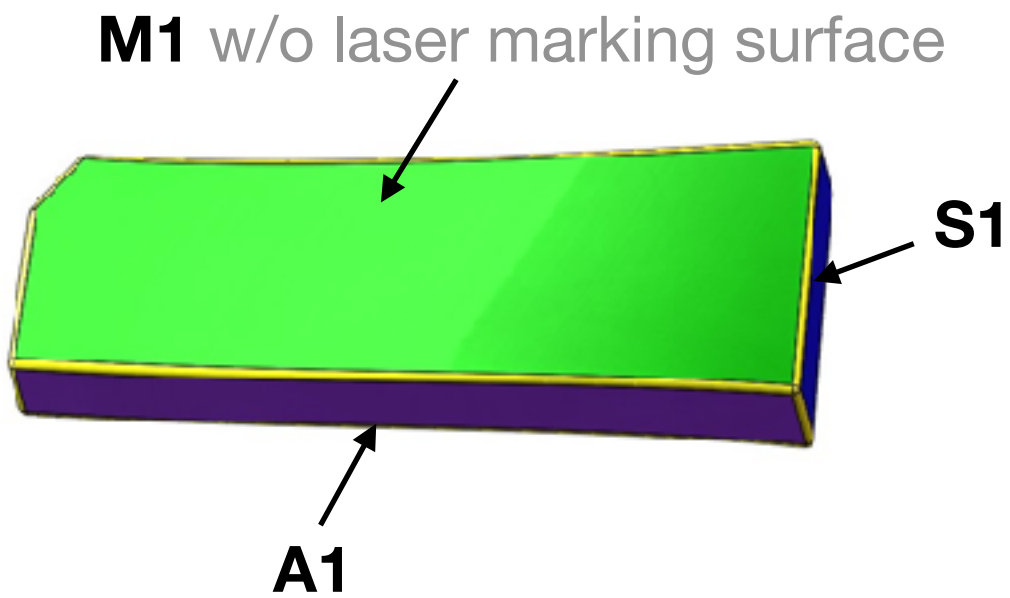
- Lighting condition: 900-1100 Lux
- Magnification not to exceed 1X unless otherwise noted
- Surface to be evaluated at a distance of 35cm
- Part to be rotated freely for inspection
- Finger Feel base on check with glove on.
- Vertical rotation angle: +/- 75° from normal (top to bottom)
- Horizontal rotation angle: +/- 75° from normal (left to right)
- View time per unit:
 - Cosmetic front surface: 3seconds
 - Cosmetic edges: 3 seconds
 - Cosmetic back side: 3seconds

Ipad Display Enclosure Cosmetic Defect Shape Template									
069-8198 Rev E									
SIZE (mm ²)	DEFECT SHAPE	CLASS	AAA	AA	A	B	C	D	E
			Outside		Hidden		Inside		
0.01		1							
0.02		2							
0.03				1					
0.05				1					
0.1					3		OK		
0.2					1	3			
0.3						1		3	
0.5								2	3
0.7								2	4
1.0									3
1.5							NG		
2.0									
2.5									
Notes	This template is to be used with A customer cosmetic spec 1. Inspection Methods (condition) 1.1 Light source: 900-1100 LUX measured at the part surface level. 1.2 Viewing distance: 35 cm 1.3 Display Angle: Part can be rotated freely to full 180 degree range 1.4 Max view time: See particular OQC document for each part. If no OQC document is available, then 10 seconds / part 2. Cosmetic 2.1 Defect types: scratch, internal contamination, black dot/line, air hole, pit, dent, crack, particle, dust, lint, fiber, sink marks, white dots, shiny marks, proud dots. 2.2 Defect Count: Surfaces classified as class AA are allowed to have 1 defect of size 0.05 mm² AND 1 defect of size 0.03 mm² provided it conforms to defect spacing rule 3.2 2.3 Rule 2.2 applies for the rest of the class types (A-E) based on the chart above. 2.4 Any defect on surface classes AA, A, B and C that breaks through the anodization layer on aluminum enclosures, regardless of its size is to be regarded as NG (the scratch or dent surface will look shiny) 3. Multiple Defect Spacing Rules 3.1 There is no spacing requirement for any one of two defects ≤ 0.02 mm². 3.2 There can be no two defects greater than 0.02mm² within 50mm distance from each other for class AA, A & B 4. Cleaning 4.1 Only A Customer approved methods to be used for cleaning of enclosure parts.								
	Contrast Standard 25% (For Class AAA, AA, A) 35% (For Class B, C, D, E) 55% Contrast standard is to be used for defects that can not be judged by the above criteria. Unless specified in cosmetic part drawing or with limit samples, use contrast standard to accept/reject for visible items such as scuff, knit lines, gate blush, gas marks, 'shock' line, shiny mark, discoloration, shadow defects etc... Scale 0 10 20 30 40 50 mm								

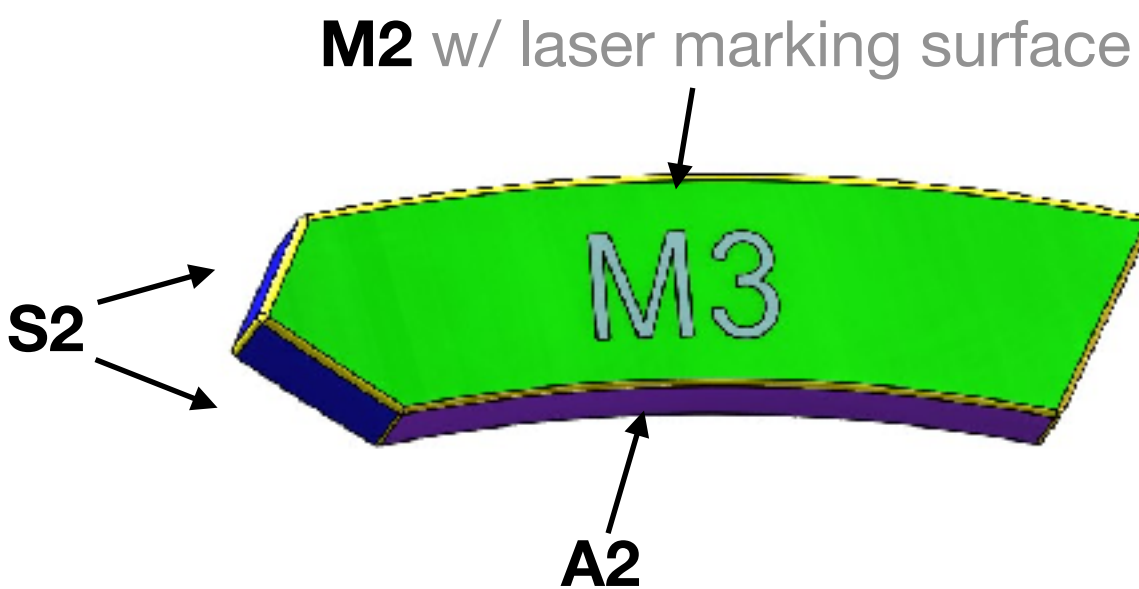
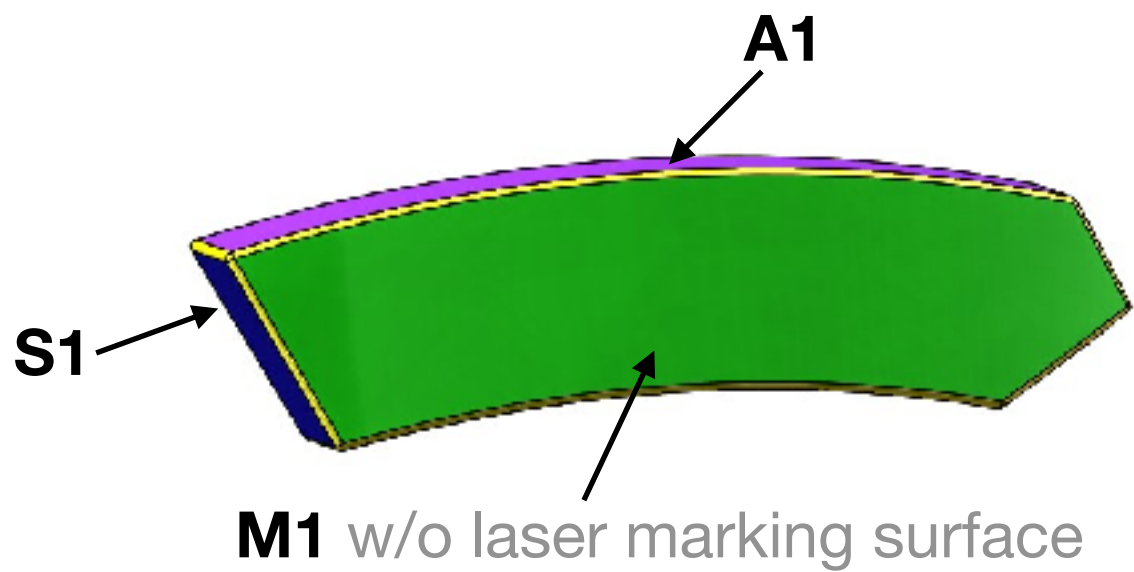
• 160-05689



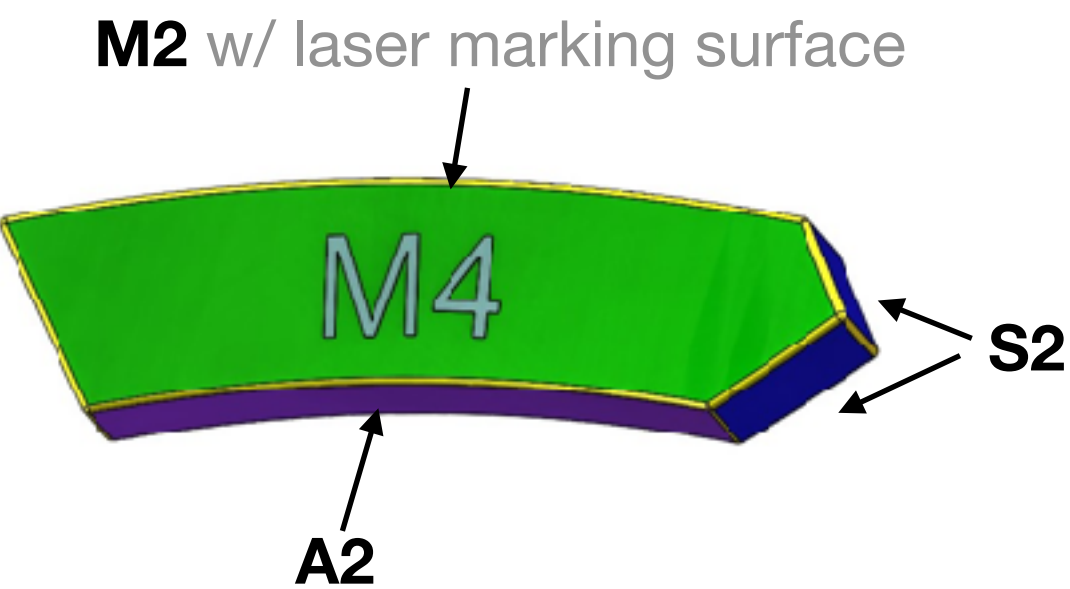
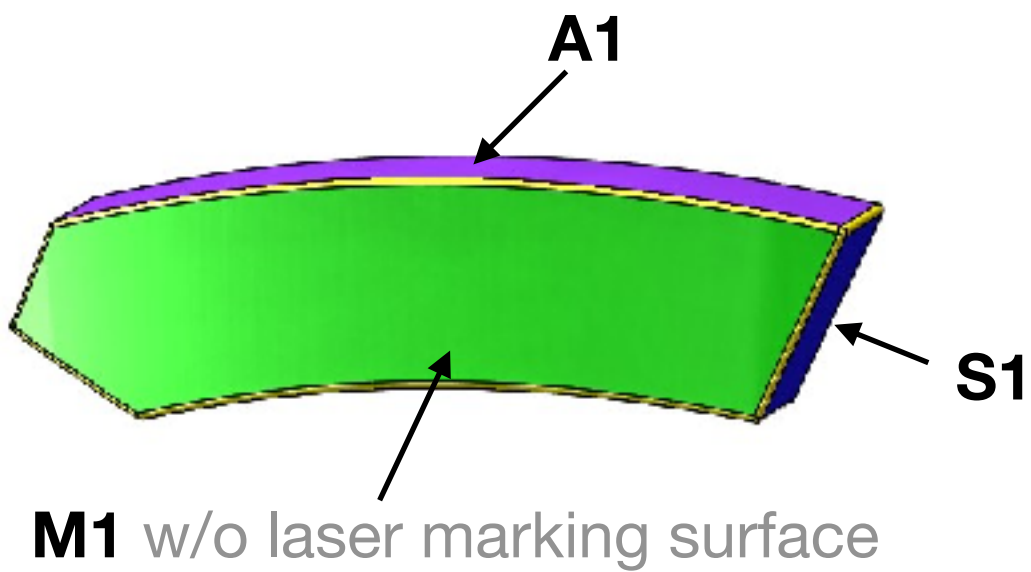
• 160-05690



• 160-05692



• 160-05691



Surface Mapping

Main Surface	Arc Surface	Edge
M1/M2	A1/A2	All edge

B747 Ramp Radial Magnet Cosmetic – CDML Index

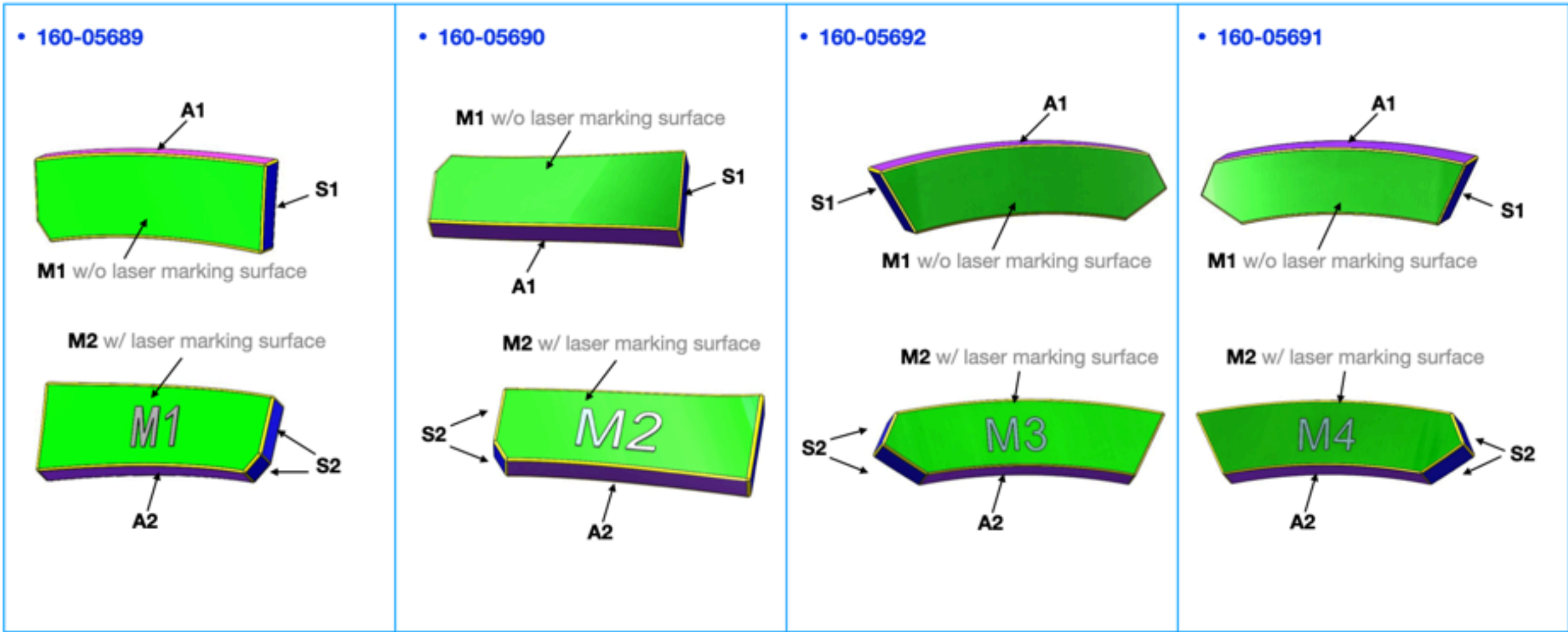
Released on:2025-08-04

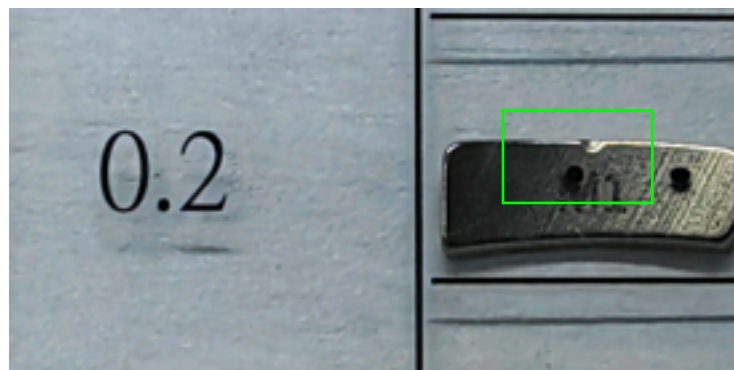
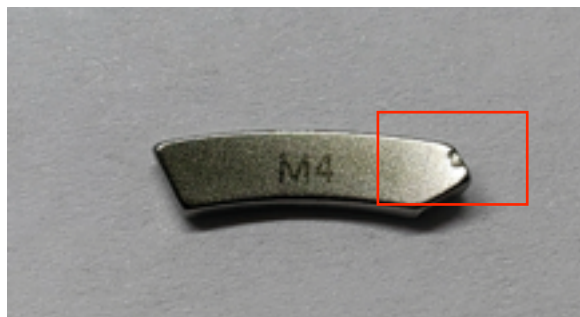
NO.	Defect Name	Location	Spec	Limit Sample Needed	APN Affected	Page #
1	Chip	All surface and Edge	Follow 069-8198 Class C 1. Individual chip w/ plating: - S>0.5mm^2: not allowed; 0.3<S≤0.5mm^2, 2x allowed per surface; 0.2<S≤0.3mm^2, 3x allowed per surface; - S≤0.2mm^2, all allowed, no qty limitation; 2. Sawtooth chip/Continuous chip w/ plating: - S<0.1mm^2, but total length<2mm, 1x allowed per edge; - S>0.1mm^2, not allowed; 3. Chip w/o plating, not allowed ;	No	160-05689 160-05690 160-05691 160-05692	6
2	Bubble	All surface and Edge	- Bubble S≤0.1mm^2, allowed, no qty limitation; - Bubble S>0.1mm^2, not allowed;	No		7
3	Scratch	All surface	1. Scratch with finger pulp feeling, not allowed ; 2. Scratch without feeling: 4mm<L≤6mm, 1x allowed per surface; 2.5mm≤L≤4mm, 3x allowed per surface; - L<2.5mm, no qty limitation.	No		8
4	Contamination	All surface	- Removable, not allowed. - Unremovable, if 35% grey film can cover, allowed; otherwise not allowed;	No		9
5	Discoloration	All surface	Cover w/ 35% grey film to check, if covered, allowed ; otherwise not allowed ;	No		10
6	Machining Mark	All surface	One line across whole surface, not allowed . Uniform lines across surface, allowed .	No		11
7	Dent/Pitting	All surface	Dent w/ plating: - S>0.2mm^2, not allowed; 0.1mm^2<S≤0.2mm^2, 3x allowed; - S≤0.1mm^2, no qty limitation; Dent w/o plating, not allowed ;	No		12
8	Plating peeling off	All surface	Plating peeling off, not allowed ;	No		13

NO.	Defect Name	Location	Spec	Limit Sample Needed	APN Affected	Page #
9	Crack	All surface	Crack, not allowed ;	No	160-05689 160-05690 160-05691 160-05692	14
10	Irregular-shape	All surface	Irregular-shaped(over cutting or missing feature), Not allowed ;	No		15
11	Laser offset	M2	Laser engraving marks are incomplete, not allowed ;	No		16
12	Lack of laser	M2	Lack of laser, not allowed .	No		17
13	Laser double shadow	M2	Laser double shadow, not allowed ;	No		18
14	Wrong Polarity	M1/M2	Wrong polarity, not allowed .	No		19
15	Weak magnetism	M1/M2	Weak magnetism, not allowed .	No		20
16	Wrong orientation	M1/M2	Wrong orientation, not allowed .	No		21

B747 Ramp Radial Magnet Cosmetic – Chip

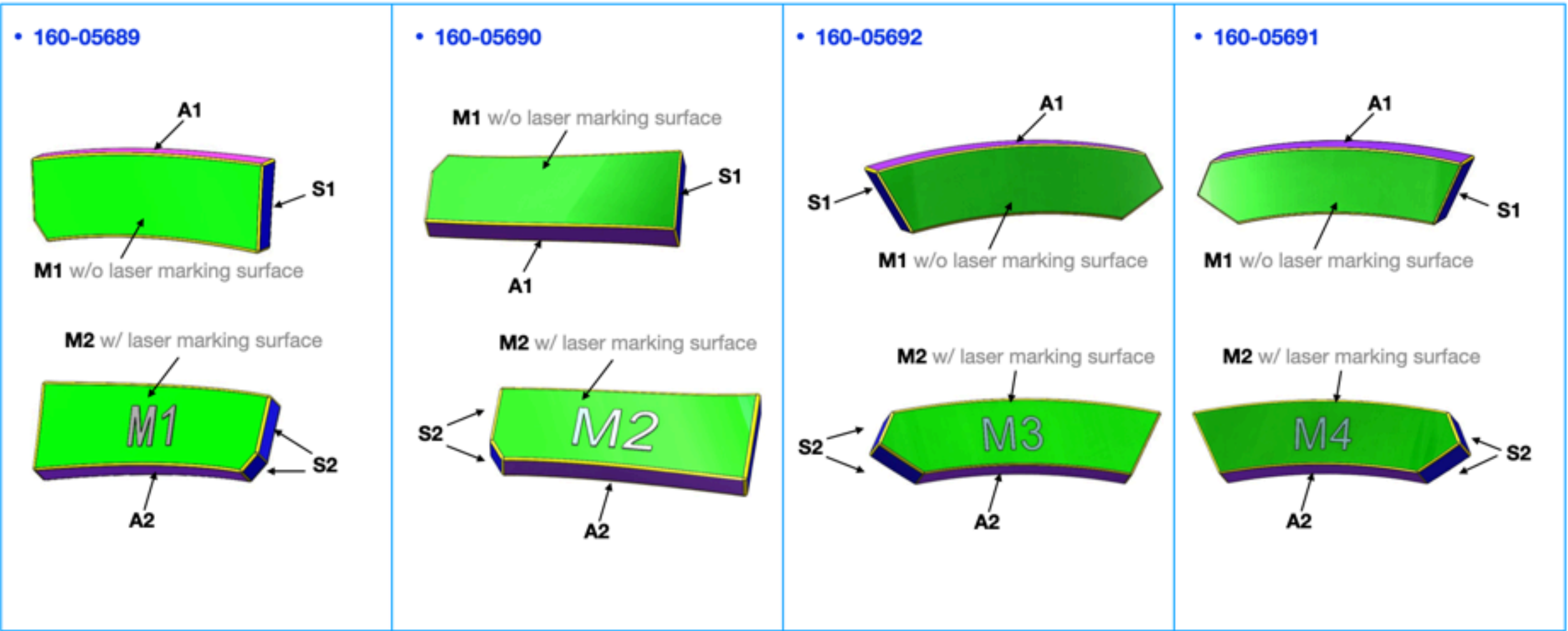
- Defect Name: Chip
- Location: All surface and Edge
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
Follow 069-8198 Class C 1. Individual chip w/ plating: • $S > 0.5\text{mm}^2$: not allowed ; $0.3 < S \leq 0.5\text{mm}^2$, 2x allowed per surface ; $0.2 < S \leq 0.3\text{mm}^2$, 3x allowed per surface ; • $S \leq 0.2\text{mm}^2$, all allowed, no qty limitation ; 2. Sawtooth chip/Continuous chip w/ plating: • $S < 0.1\text{mm}^2$, but total length $< 2\text{mm}$, 1x allowed per edge ; $S > 0.1\text{mm}^2$, not allowed ; 3. Chip w/o plating, not allowed ;	Moderate	Main Surface	Arc Surface	Side Surface	Edge
		M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 $S \leq 0.2\text{mm}^2$, OK  $S \leq 0.3\text{mm}^2$, OK	/	 $S > 0.5\text{mm}^2$, not allowed. 			

B747 Ramp Radial Magnet Cosmetic – Bubble

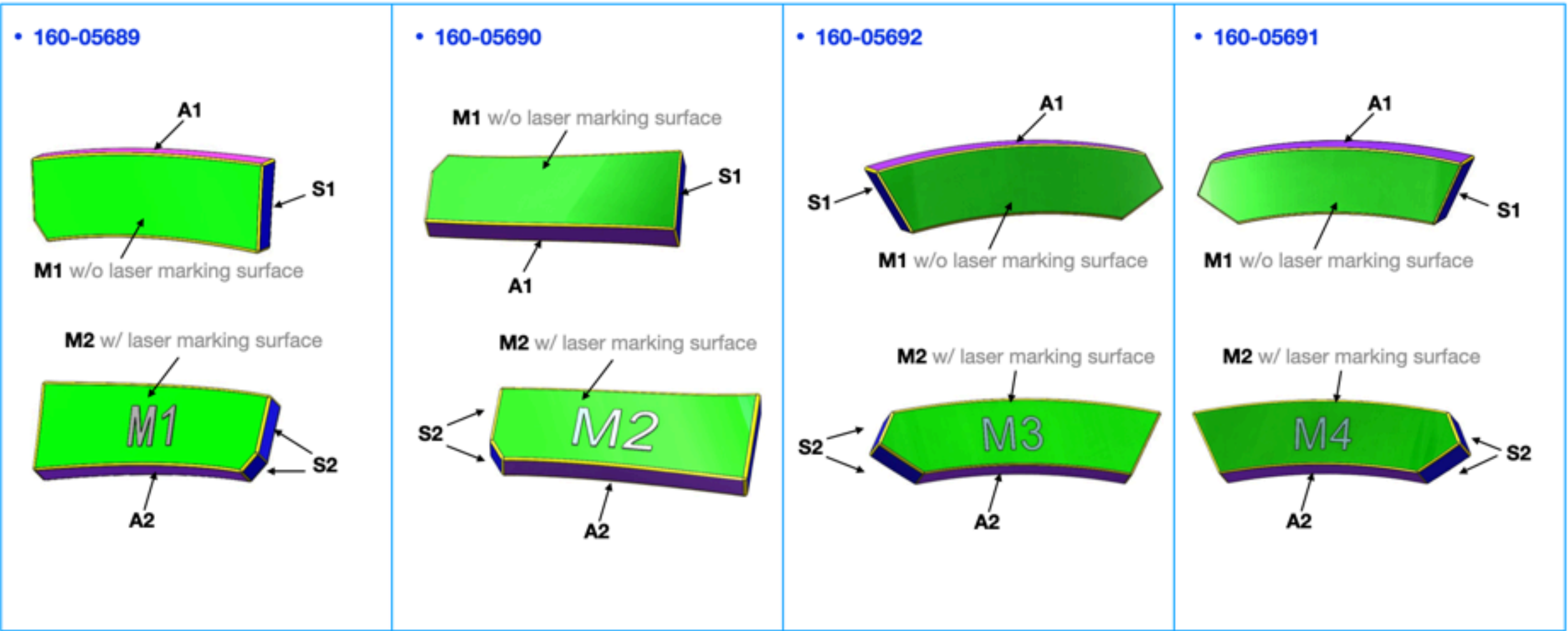
- Defect Name: Bubble
- Location: All surface and Edge
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
• Bubble $S \leq 0.1\text{mm}^2$, allowed, no qty limitation; • Bubble $S > 0.1\text{mm}^2$, not allowed;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
Bubble $S \leq 0.1\text{mm}^2$, allowed $S \leq 0.1\text{mm}^2$	/	$S > 0.1\text{mm}^2$, not allowed			

B747 Ramp Radial Magnet Cosmetic – Scratch

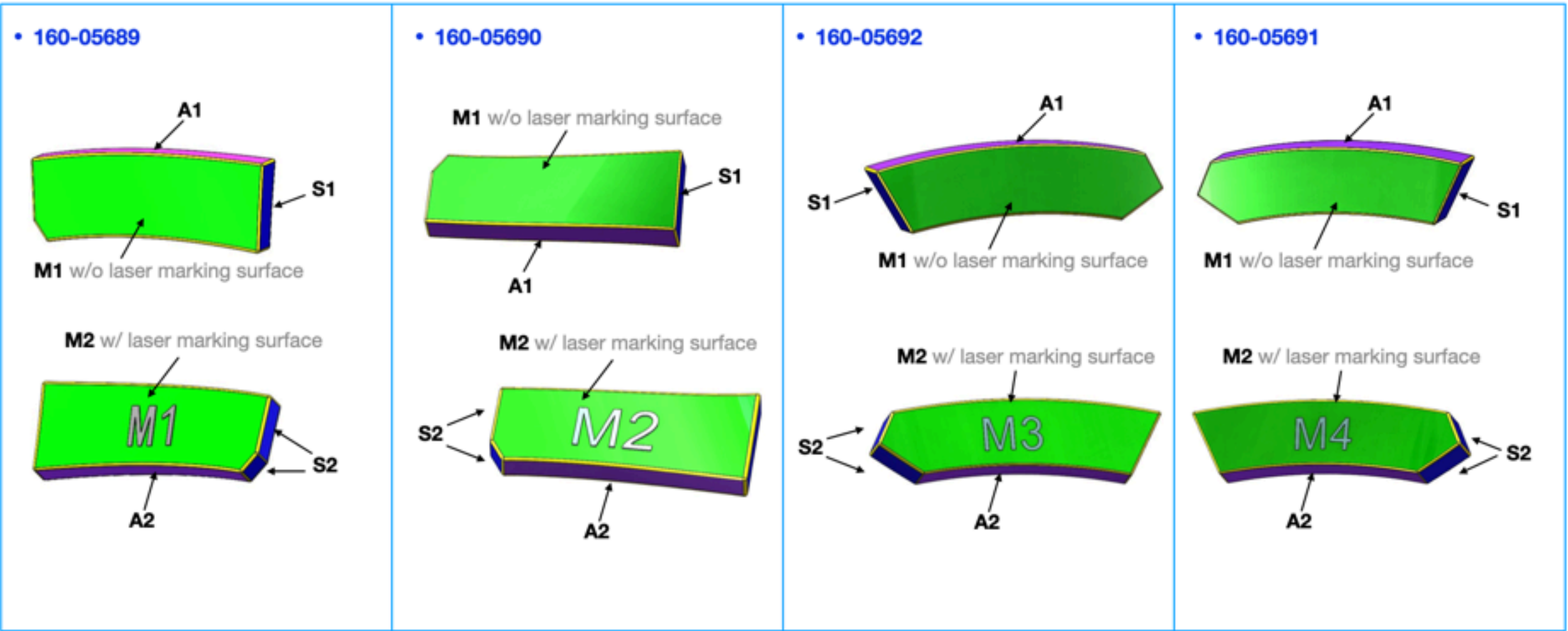
- Defect Name: Scratch
- Location: All surface
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
1. Scratch with finger pulp feeling, not allowed ; 2. Scratch without feeling: • 4mm<L≤6mm, 1x allowed per surface; • 2.5mm≤L≤4mm, 3x allowed per surface; • L<2.5mm, no qty limitation.	Moderate	Main Surface	Arc Surface	Side Surface	Edge
		M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 L≤4mm, allowed	/	 L>6mm, not allowed  Scratch with finger pulp feeling, not allowed			

B747 Ramp Radial Magnet Cosmetic – Contamination

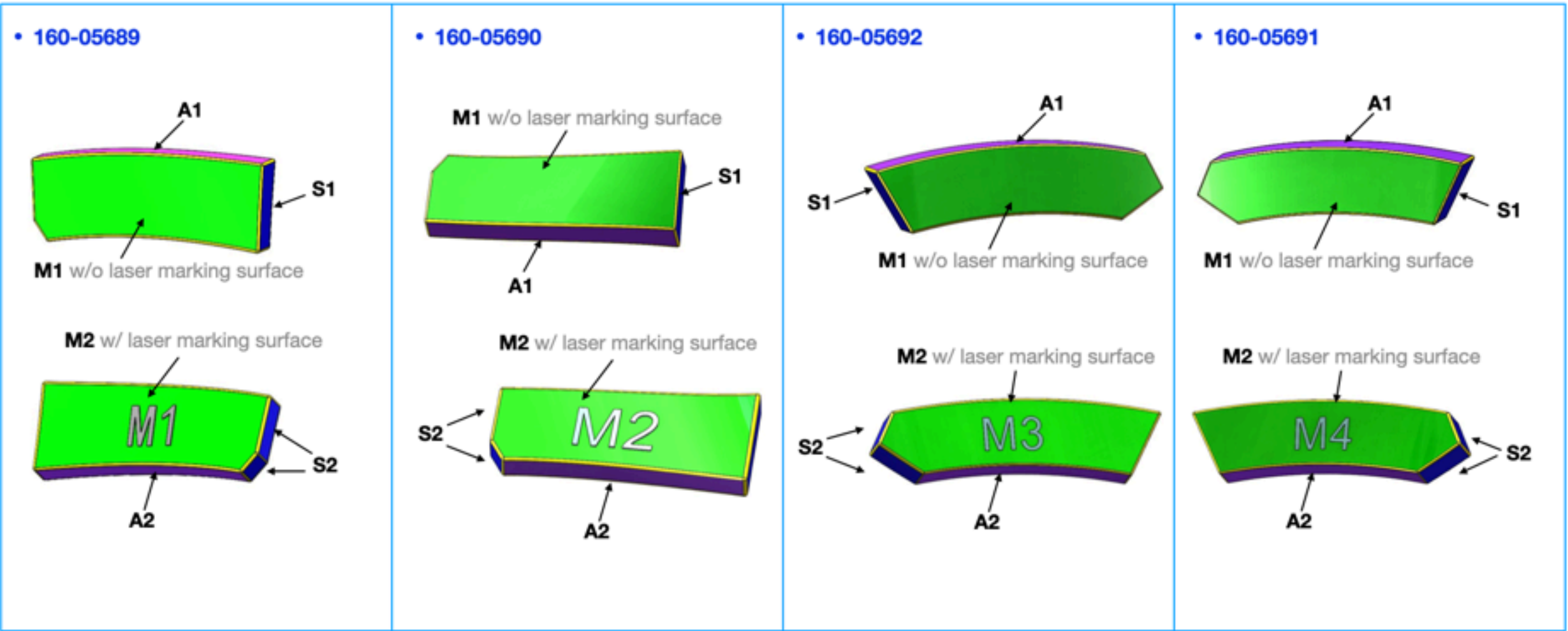
- Defect Name: Contamination
- Location: All surface
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
• Removable, not allowed. • Unremovable, if 35% grey film can cover, allowed; otherwise not allowed;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 35% (For Classes B,C,D) Unremovable, if 35% grey film can cover, allowed;	/	/			

B747 Ramp Radial Magnet Cosmetic – Discoloration

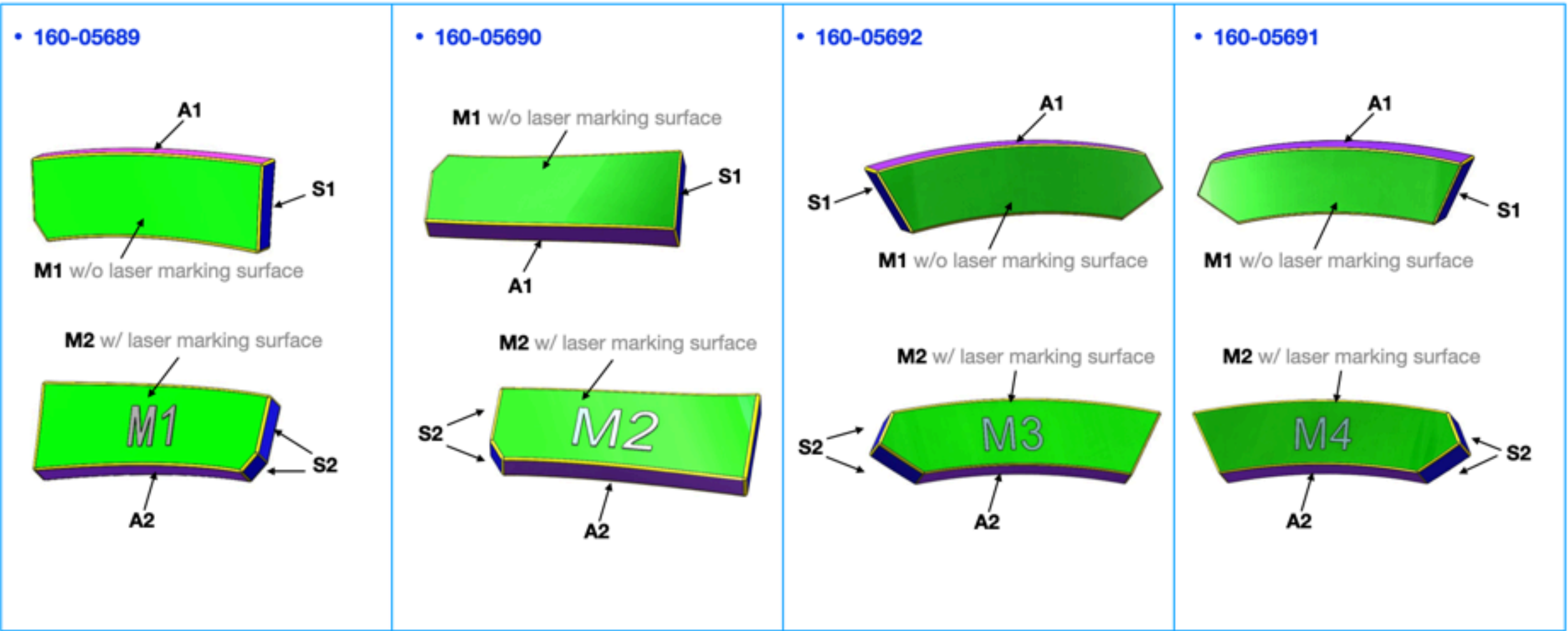
- Defect Name: **Discoloration**
- Location: **All surface**
- APN Affected: **160-05689/160-05690/160-05691/160-05692**

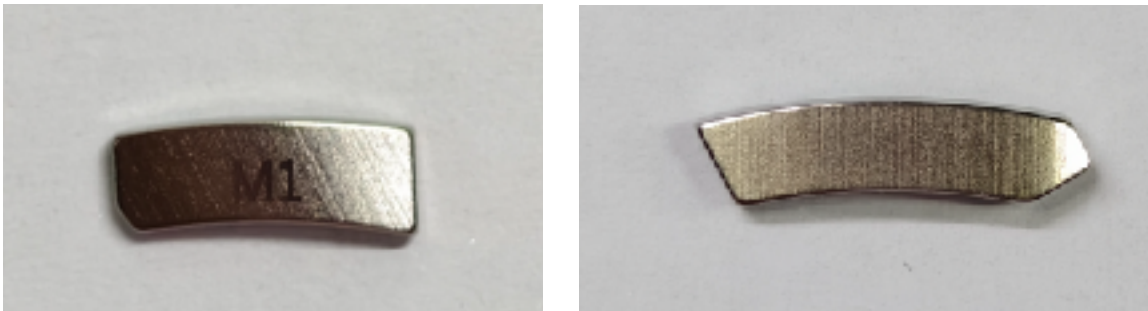


Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
• Cover w/ 35% grey film to check, if covered, allowed; otherwise not allowed;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
/	/	/			

B747 Ramp Radial Magnet Cosmetic – Machining Mark

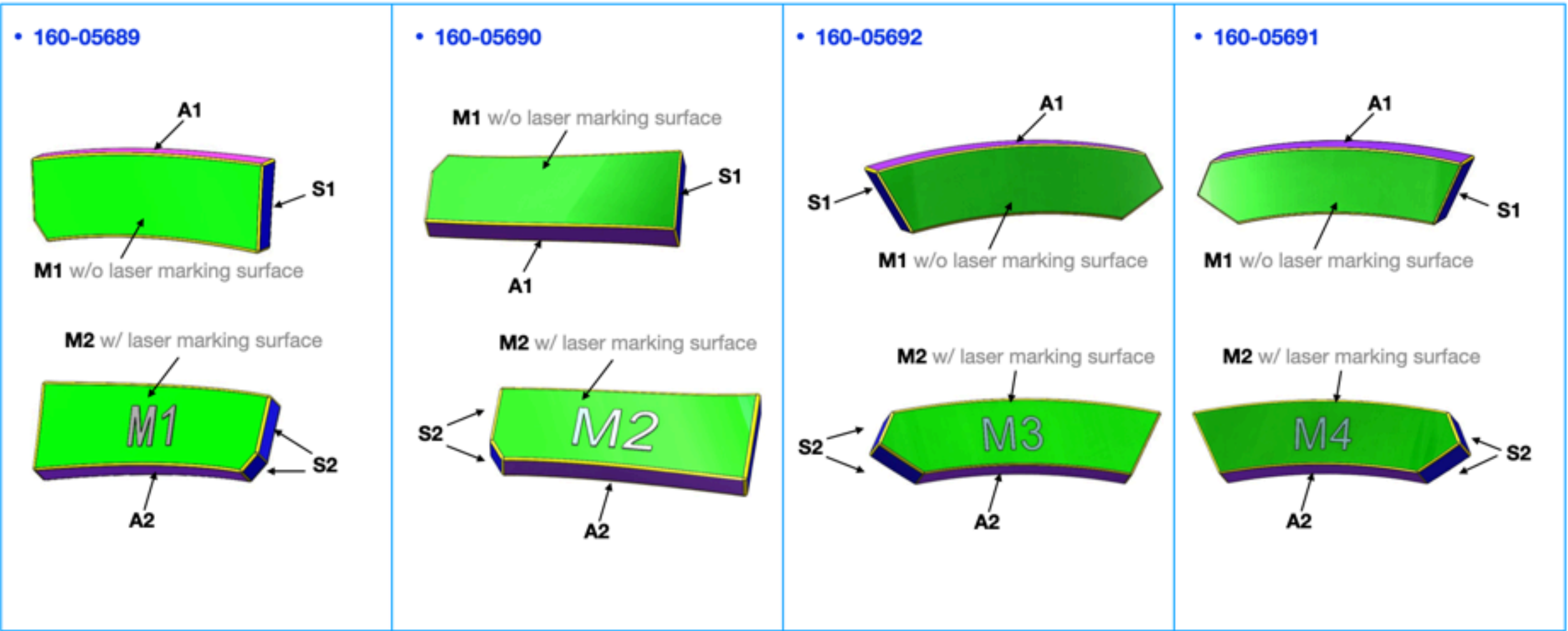
- Defect Name: Scratch
- Location: All surface
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
One line across whole surface, not allowed . Uniform lines across surface, allowed .	Moderate	Main Surface	Arc Surface	Side Surface	Edge
		M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 Uniform lines across surface, allowed	/	 W/O feeling: One line across whole surface, not allowed			

B747 Ramp Radial Magnet Cosmetic – Dent/Pitting

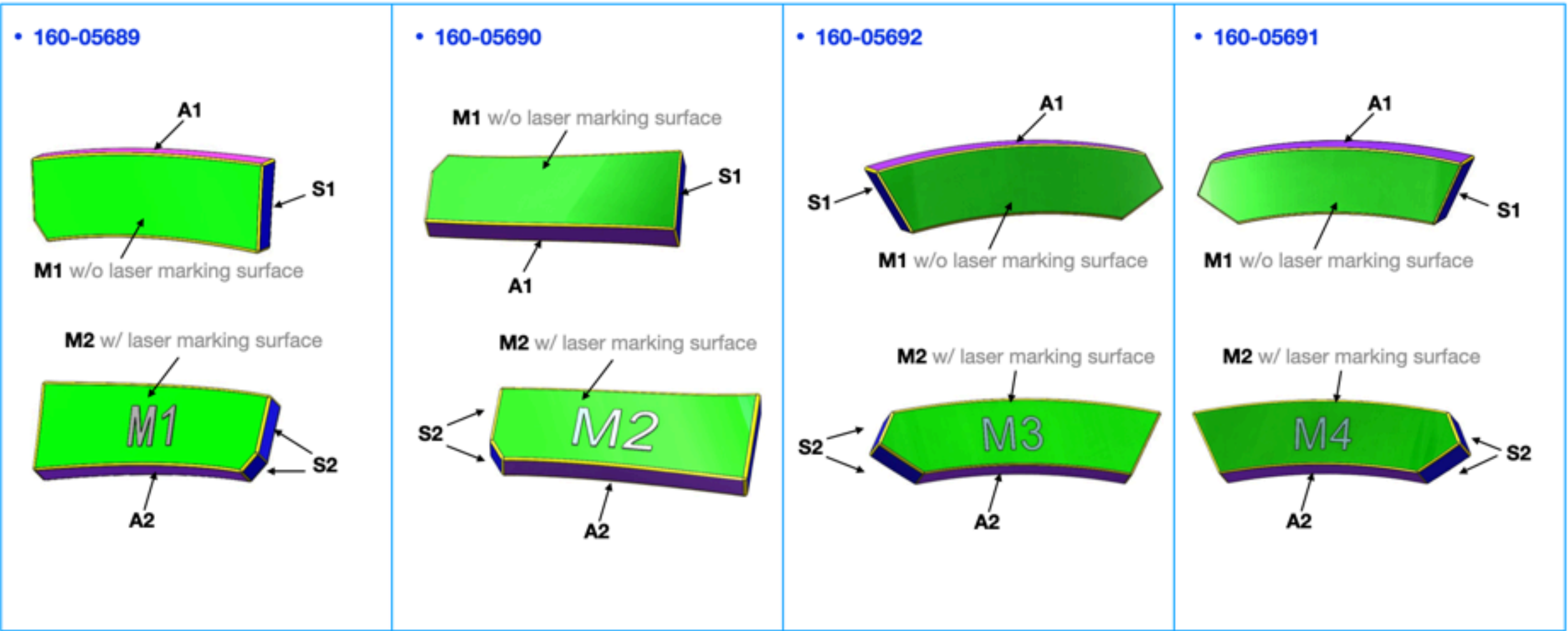
- Defect Name: Dent/Pitting
- Location: All surface
- APN Affected: 160-05689/160-05690/160-05691/160-05692

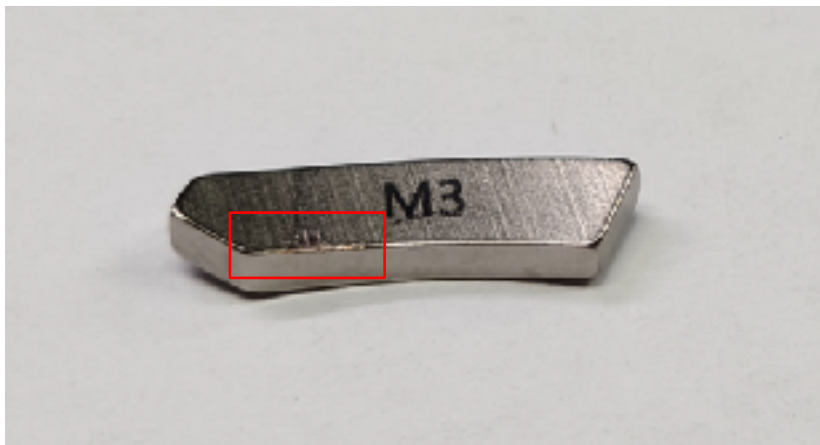


Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Dent w/ plating: • $S > 0.2\text{mm}^2$, not allowed; • $0.1\text{mm}^2 < S \leq 0.2\text{mm}^2$, 3x allowed; • $S \leq 0.1\text{mm}^2$, no qty limitation; Dent w/o plating, not allowed;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 0.1mm² < S ≤ 0.2mm², 3x allowed;	/	 4x, not allowed			

B747 Ramp Radial Magnet Cosmetic – Plating peeling off

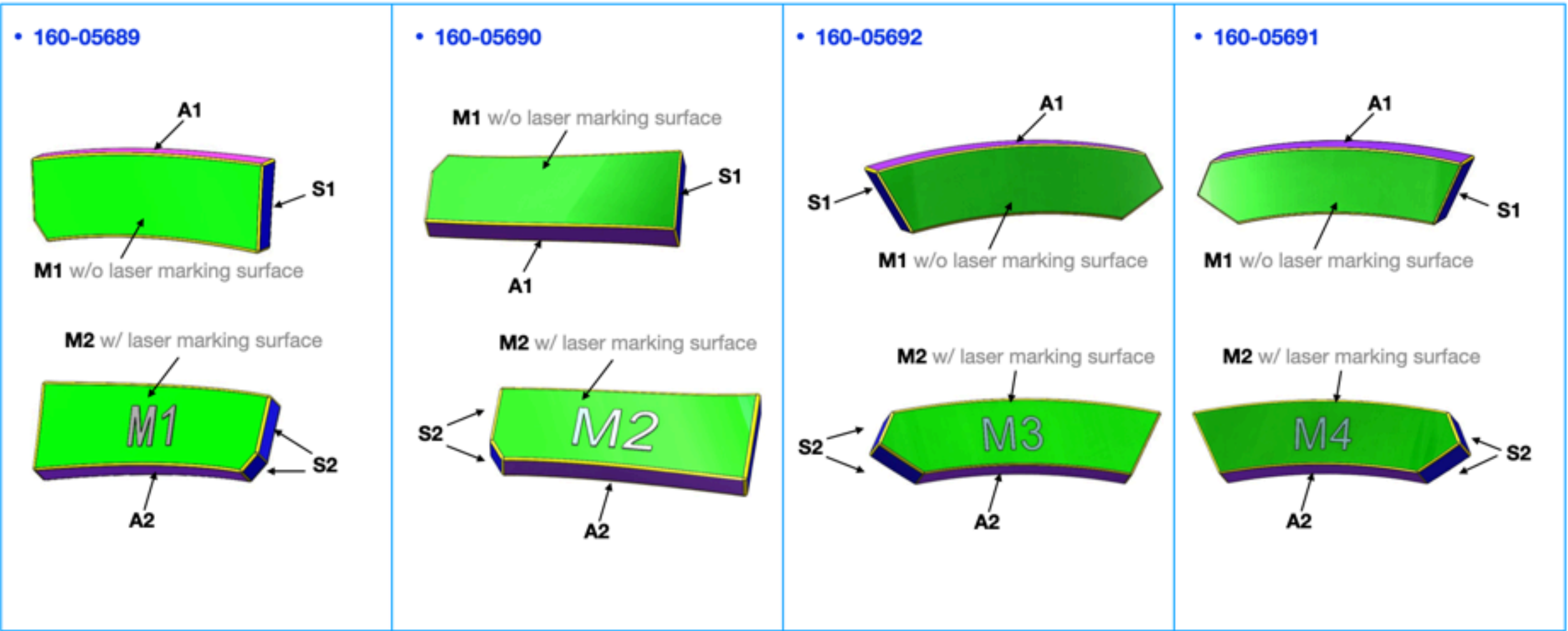
- Defect Name: **Plating peeling off**
- Location: **All surface**
- APN Affected: **160-05689/160-05690/160-05691/160-05692**

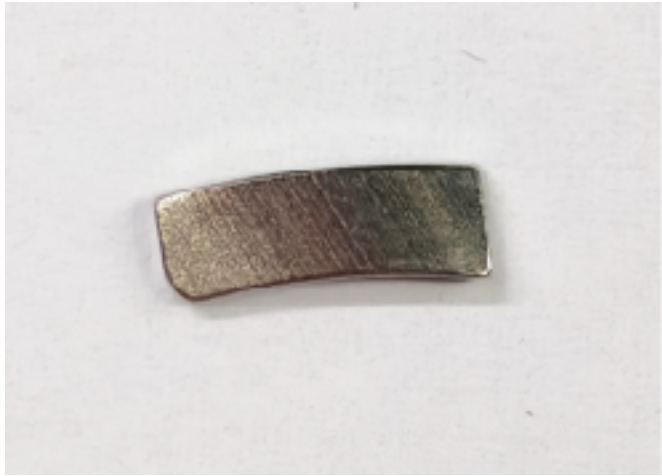


Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Plating peeling off, not allowed ;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 OK Sample	/	 Dent w/o plating, not allowed;			

B747 Ramp Radial Magnet Cosmetic –Crack

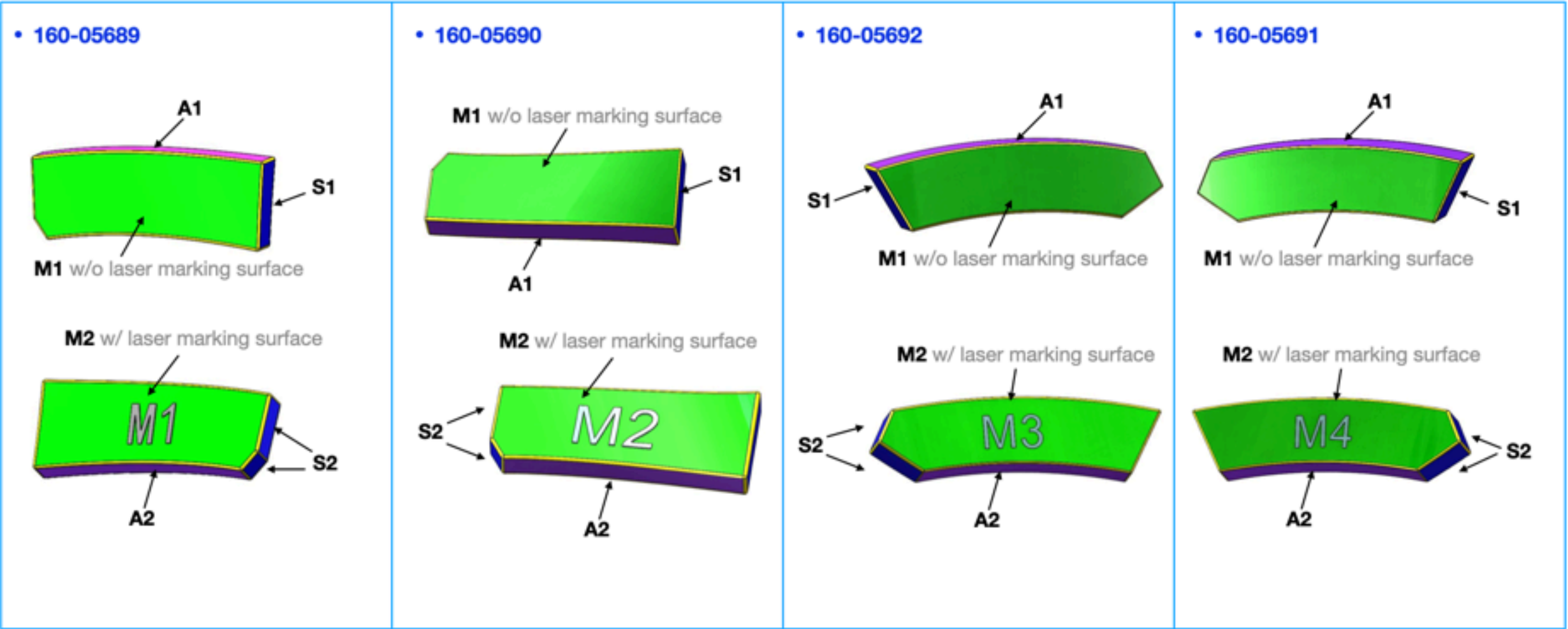
- Defect Name:Crack
- Location: All surface
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Crack, not allowed ;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 OK Sample	/	 Crack, not allowed;			

B747 Ramp Radial Magnet Cosmetic –Irregular-shape

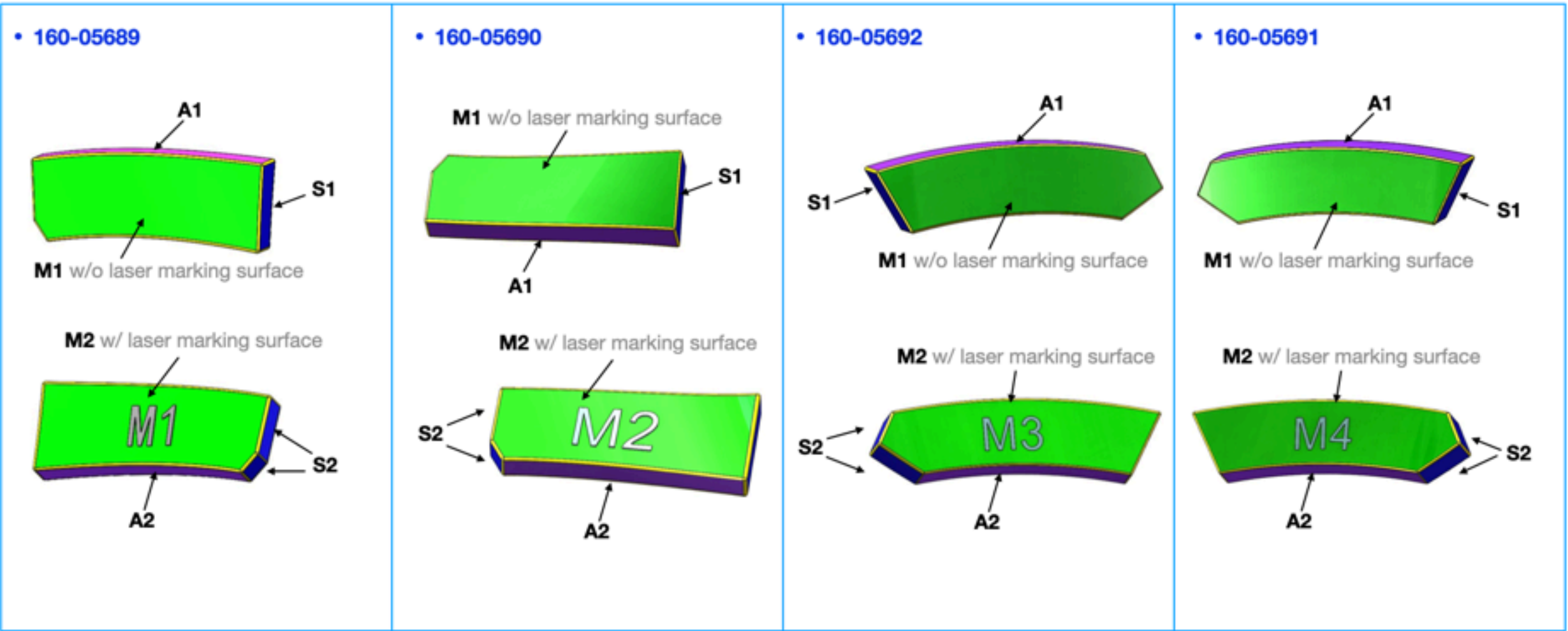
- Defect Name:Irregular-shape
- Location: All surface
- APN Affected: 160-05689/160-05690/160-05691/160-05692

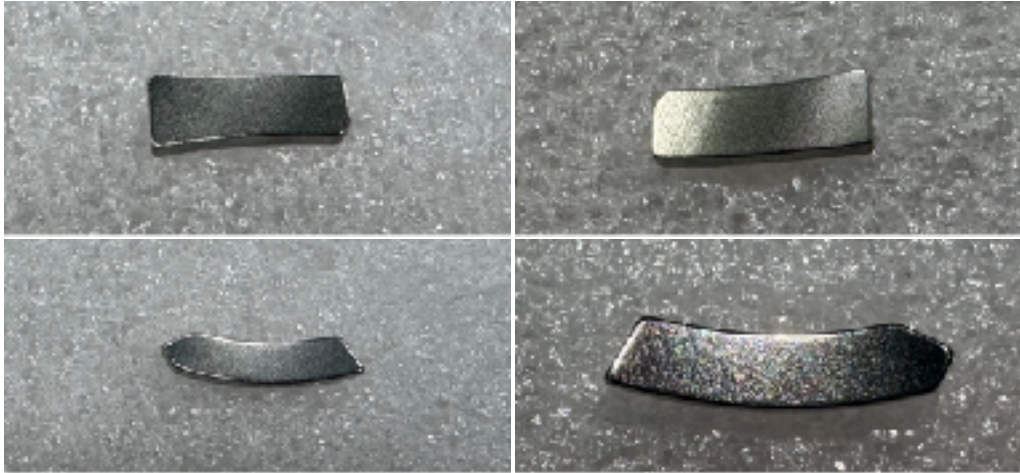


Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Irregular-shaped(over cutting or missing feature), Not allowed;	Moderate	M1/M2	A1/A2	S1/S2	All edge
		NG			
OK	/				
OK Sample	/				
		Irregular-shaped(over cutting or missing feature), Not allowed;			

B747 Ramp Radial Magnet Cosmetic – Laser offset

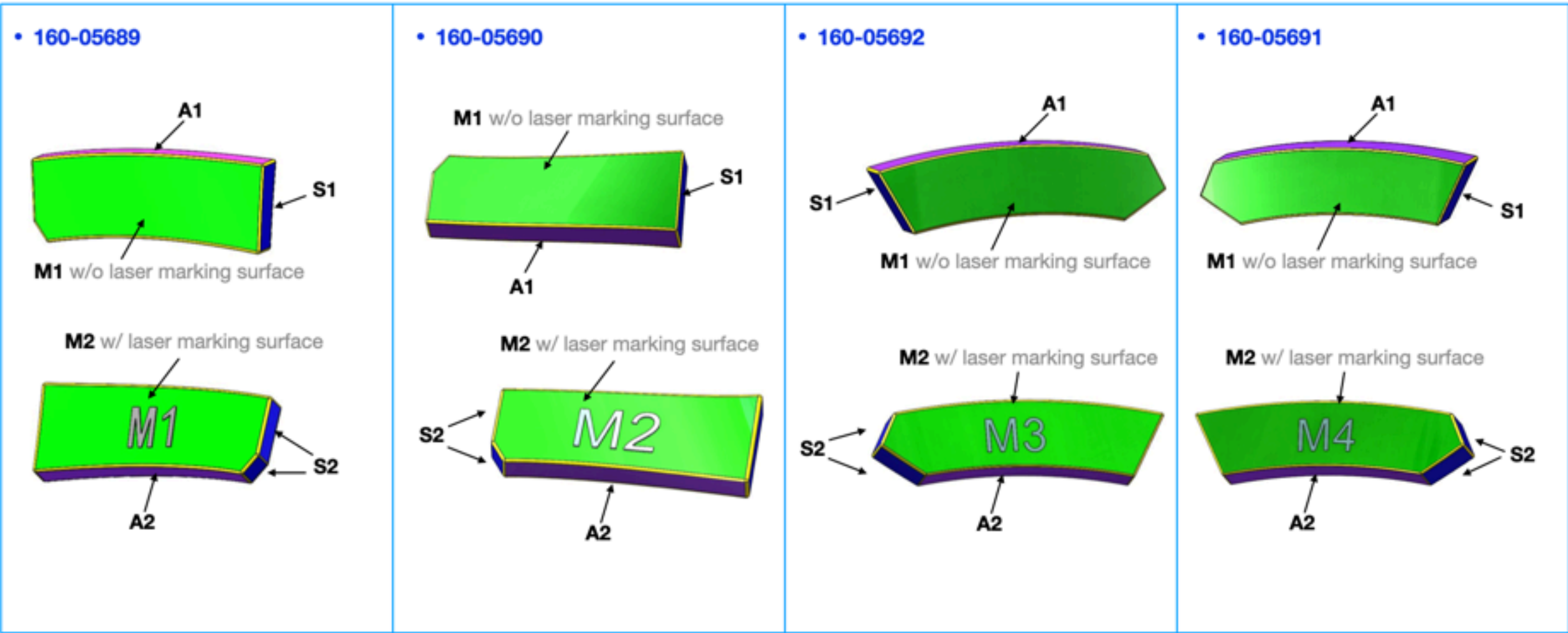
- Defect Name: Laser offset
- Location: M2
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Laser engraving marks are incomplete, not allowed ;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 OK Sample	/	 Laser engraving marks are incomplete, not allowed;			

B747 Ramp Radial Magnet Cosmetic – Lack of laser

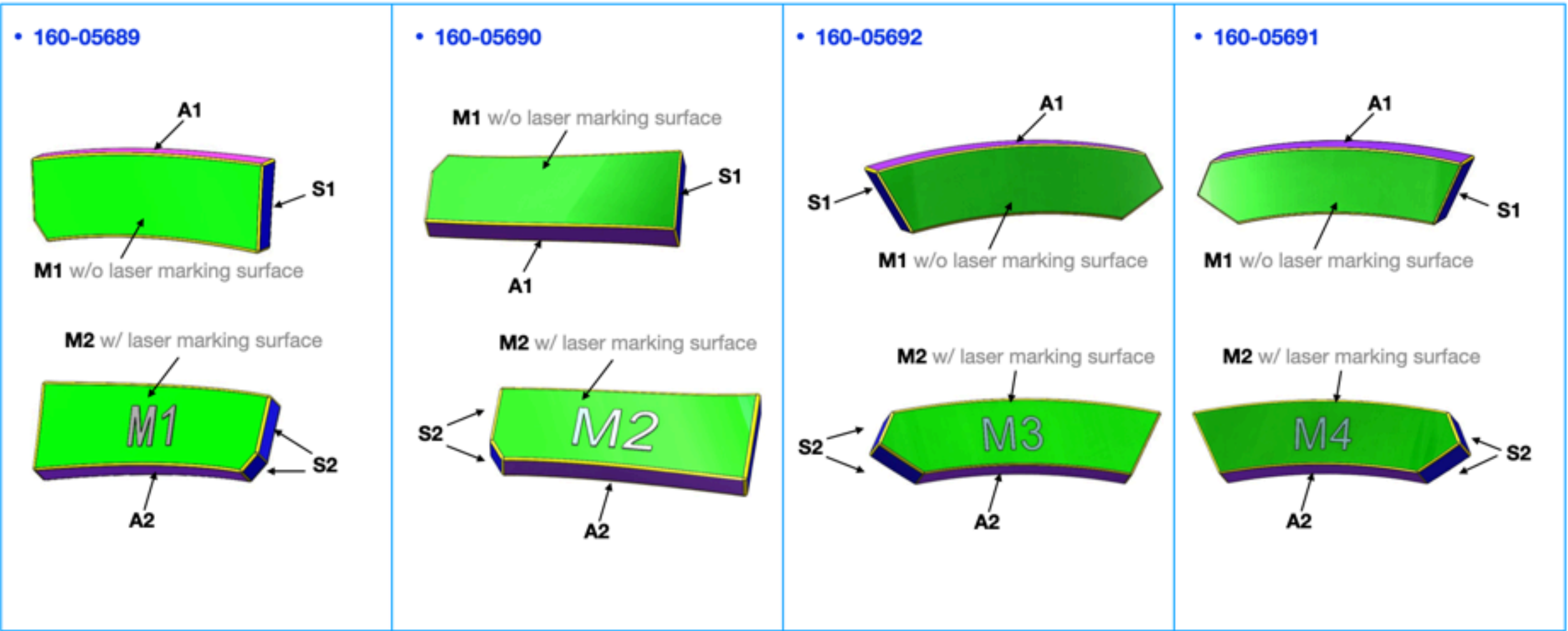
- Defect Name: Lack of laser
- Location: M2
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Lack of laser, not allowed.	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 OK Sample	/	 Lack of laser, not allowed.			

B747 Ramp Radial Magnet Cosmetic – Laser double shadow

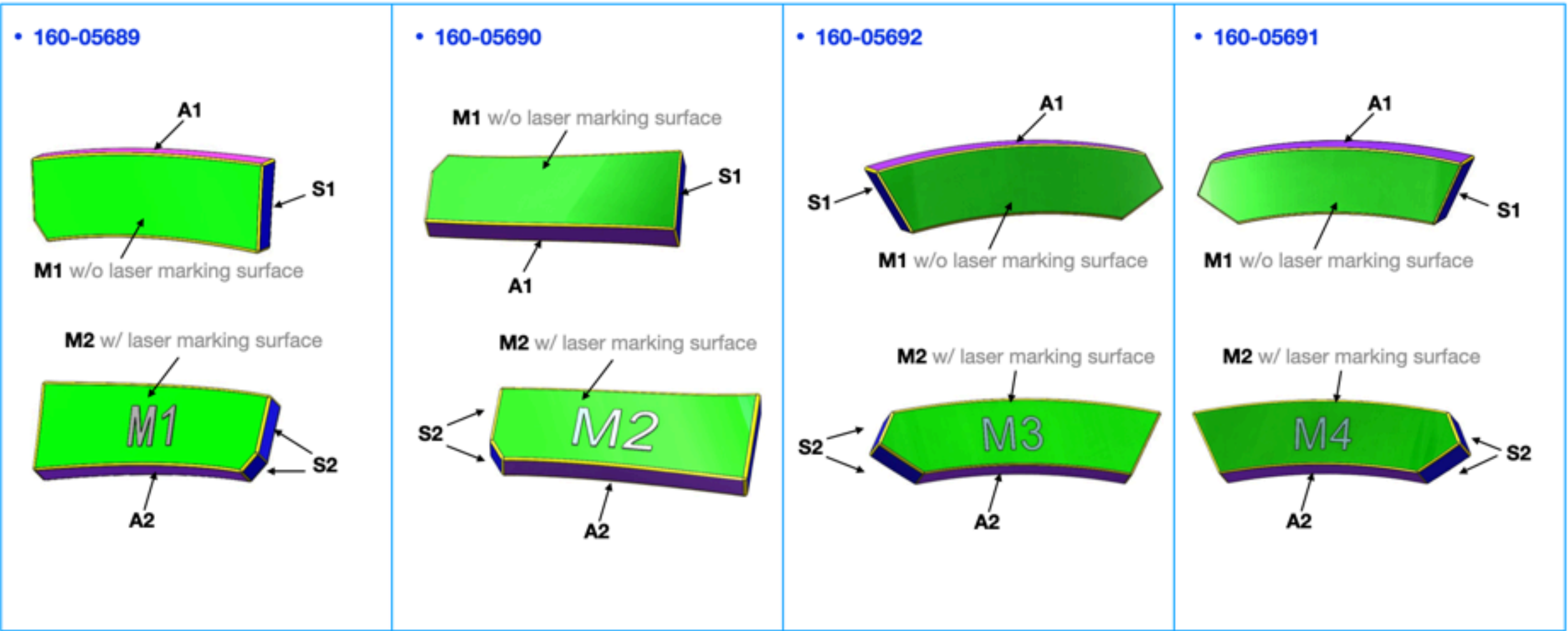
- Defect Name: Laser double shadow
- Location: M2
- APN Affected: 160-05689/160-05690/160-05691/160-05692

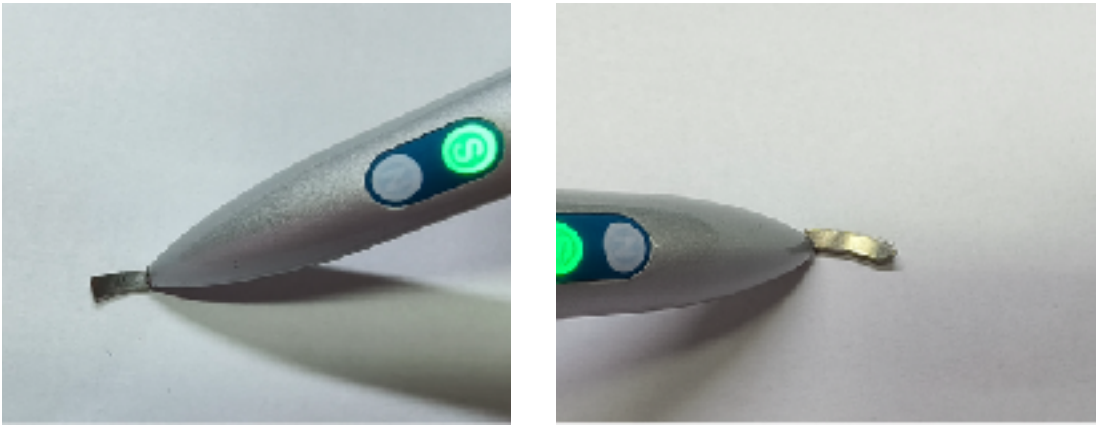
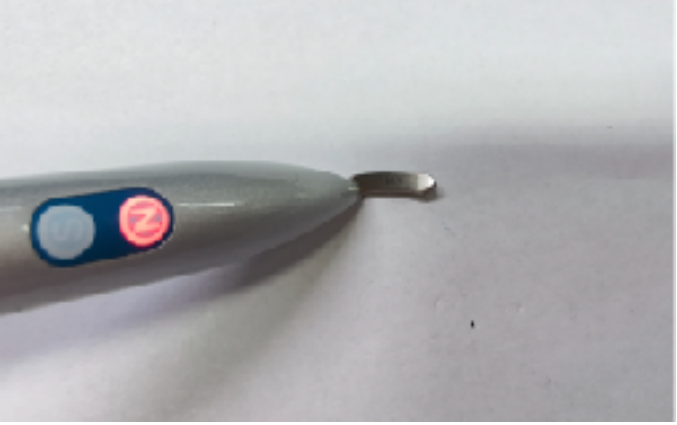


Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Laser double shadow, not allowed ;	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 OK Sample	/	 Laser double shadow, not allowed;			

B747 Ramp Radial Magnet Cosmetic – Wrong Polarity

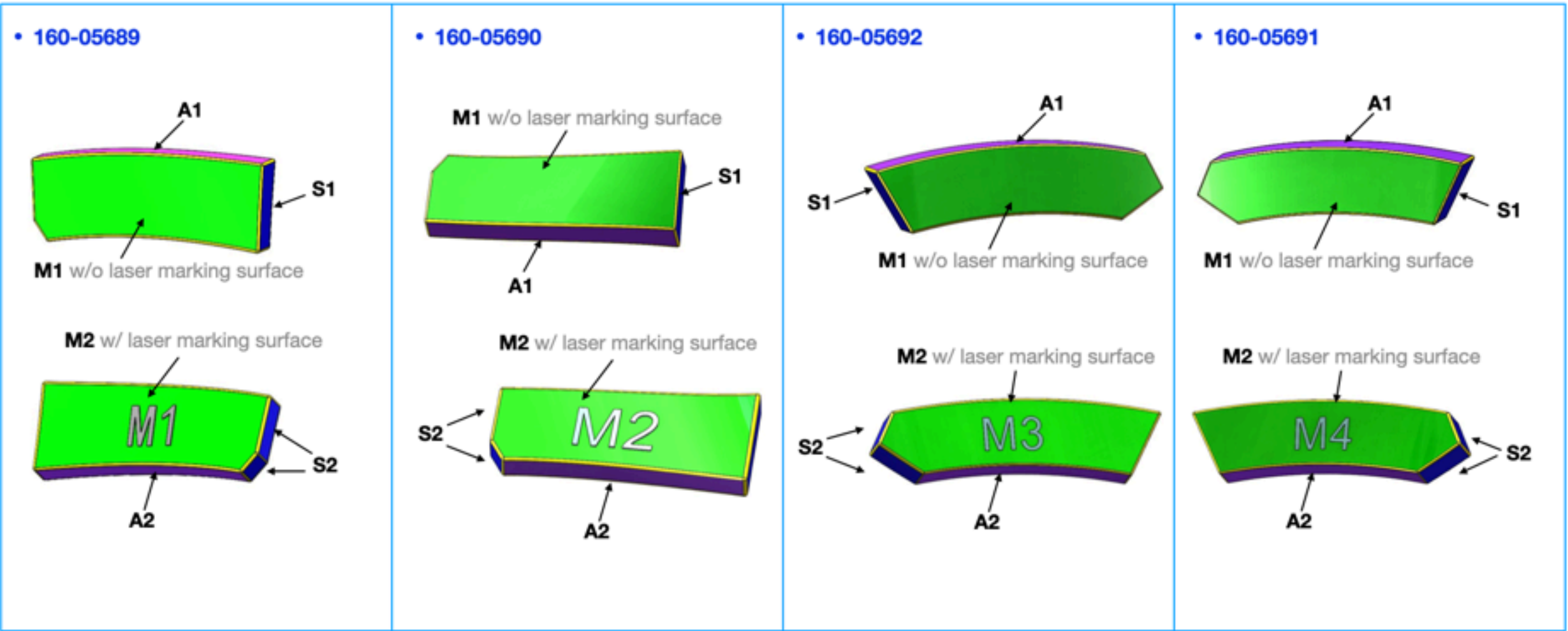
- Defect Name: Wrong Polarity
- Location: M1/M2
- APN Affected: 160-05689/160-05690/160-05691/160-05692



Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Wrong polarity, not allowed .	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
 M2 -“S”. Wrong polarity, OK	/	 M2 -“N”. Wrong polarity, not allowed.			

B747 Ramp Radial Magnet Cosmetic – Weak magnetism

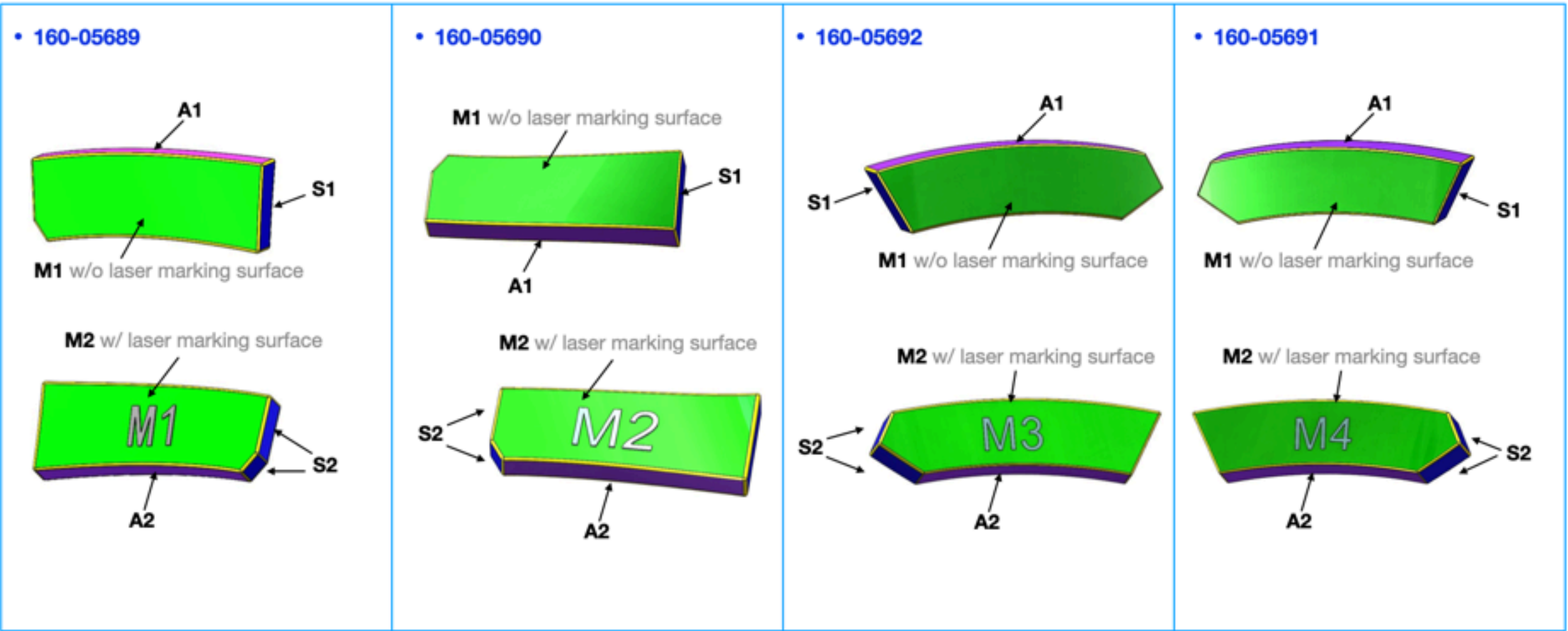
- Defect Name: Weak magnetism
- Location: M1/M2
- APN Affected: 160-05689/160-05690/160-05691/160-05692



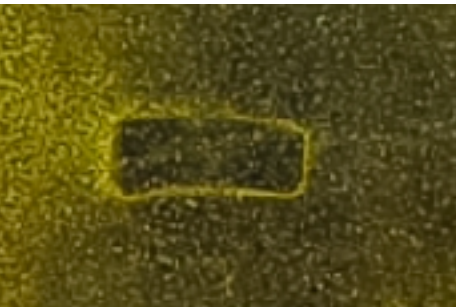
Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Weak magnetism, not allowed .	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
M2 (laser Mark) face up test OK samples	/	Wrong polarity, not allowed.			

B747 Ramp Radial Magnet Cosmetic –Wrong orientation

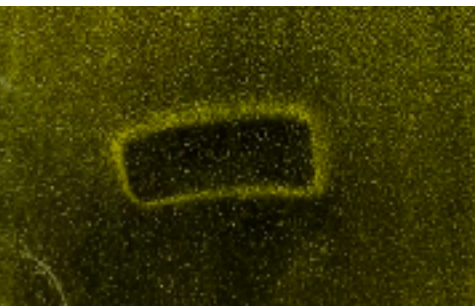
- Defect Name: Wrong orientation
- Location: M1/M2
- APN Affected: 160-05689/160-05690/160-05691/160-05692



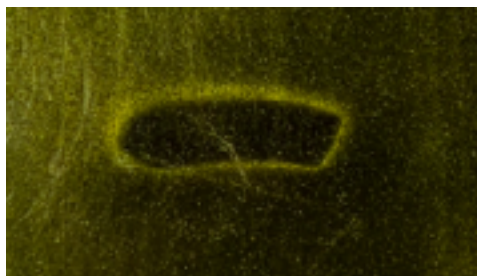
Spec	Alert Level	Surface Mapping			
		Main Surface	Arc Surface	Side Surface	Edge
Wrong orientation, not allowed.	Moderate	M1/M2	A1/A2	S1/S2	All edge
OK	/	NG			
M2 (laser Mark) face up test OK samples	/	/			



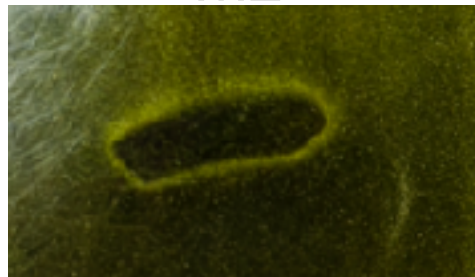
M1



M2



M3



M4